**Unit:** 6 Unit 9: Data Statistical Analysis **Name:** ____________
**Topic:** Box Whisker (Dispersion) **Date:** ________

# Advanced (Habanero)

**Q1.** What does the box in a box-and-whisker plot represent?
(A) The range of the data set
(B) The interquartile range
(C) The middle 50
(C) The mean of the data set
(D) The median of the data set

**Q2.** What is the primary purpose of the whiskers in a box-and-whisker plot?
(A) To show the mean and median
(B) To show the range of the data
(C) To identify outliers
(D) To display the mode
(E) To illustrate the standard deviation

**Q3.** If $Q_1 = 12$ and $Q_3 = 20$, what is the interquartile range ($IQR$)?
(A) 8
(B) 12
(C) 16
(D) 20
(E) 24

**Q4.** Which of the following is an outlier in the data set $1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 100$?
(A) 1
(B) 5
(C) 10
(D) 100
(E) None of the above

**Q5.** A box-and-whisker plot shows $Q_1 = 15$, $Q_3 = 25$, and the lowest value is 5. What is the range of the data set?
(A) 10
(B) 20
(C) 30
(D) 40
(E) 50

**Q6.** What percentage of the data falls within the interquartile range ($IQR$)?
(A) 25
(A) 50
(A) 75
(A) 90
(A) 100

**Q7.** The median of a data set is 18, $Q_1 = 12$, and $Q_3 = 22$. What can be said about the distribution of the data?
(A) It is symmetric
(B) It is skewed to the left
(C) It is skewed to the right
(D) It is bimodal
(E) Insufficient information

**Q8.** In a box-and-whisker plot, what does the line inside the box represent?
(A) The mean
(B) The median
(C) The mode
(D) The range
(E) The interquartile range

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**Open Ended Questions (FRQ)**

**Q9.** Construct a box-and-whisker plot for the data set $2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30$. Identify $Q_1$, $Q_3$, and the interquartile range ($IQR$).

**Q10.** Explain how to detect outliers using a box-and-whisker plot. Provide an example with a data set of your choice.

**Chef's Tips**



- Tip 1: Ensure students understand the concept of quartiles before moving to box-and-whisker plots.
- Tip 2: Use real-world data sets to make the concepts more relatable and interesting.
- Tip 3: Encourage students to analyze and interpret the results of their box-and-whisker plots.